API Documentation: AI-Powered Real Estate Property Valuation Assistant

Base URL: `http://localhost:8000`

API Prefix: `/api/v1`

Interactive OpenAPI UI: `http://localhost:8000/docs`

1. System Health Check

- **Endpoint**: `/health`
- **HTTP Method**: `GET`
- **Description**: Returns microservice operational status.
- **Request Format**: None
- **Sample Request**:

```
curl -X GET http://localhost:8000/health
```

- **Sample Response**:

```
{
"status": "healthy",
"service": "AI-Powered Real Estate Property Valuation Assistant",
"version": "1.0.0"
}
```

2. Input Validation Guardrails

- **Endpoint**: `/api/v1/guardrails/validate`
- **HTTP Method**: `POST`
- **Description**: Validates structured property features and natural-language user queries against physical boundaries, out-of-bounds parameters, and prompt injection attacks.
- **Request Format**:

```
{
"features": {
"gr_liv_area": 1800,
"bedrooms": 3,
"bathrooms": 2,
"year_built": 1995,
"overall_qual": 7
},
"query": "3 bedroom house in North Ames"
}
```

- **Sample Response**:

```
{
"is_valid": true,
"sanitized_input": {
"gr_liv_area": 1800.0,
"bedrooms": 3,
"bathrooms": 2.0,
"year_built": 1995,
"overall_qual": 7
},
"errors": [],
"warnings": []
}
```

3. Hybrid Search & Comparable-Property Prioritization

- **Endpoint**: `/api/v1/search`
- **HTTP Method**: `POST`

• **Description**: Executes true hybrid retrieval combining dense semantic embeddings (`all-MiniLM-L6-v2`) and lexical TF-IDF keyword search over the full filtered candidate set. Supports metadata filtering (including `min_total_rooms` and `max_total_rooms`) and multi-factor comparable-property prioritization ranking.

• **Empty Filter Behavior**: When metadata filters match 0 properties, returns an empty list (`[]`) with `count: 0` instead of falling back to all properties.

• **Validation**: Rejects invalid natural language queries or invalid `target_property` features with HTTP `400 Bad Request`.

• **Request Format**:


```
{
"query": "3 bedroom single family home in North Ames built after 1990",
"top_k": 3,
"filters": {
"neighborhood": "North Ames",
"min_bedrooms": 3,
"min_total_rooms": 6,
"max_total_rooms": 10
},
"target_property": {
"neighborhood": "North Ames",
"gr_liv_area": 1800,
"bedrooms": 3,
"bathrooms": 2,
"year_built": 1995,
"overall_qual": 7,
"total_rooms": 7,
"garage_yr_blt": 1995
}
}
```


• **Sample Response**:


```
{
"query": "3 bedroom single family home in North Ames built after 1990",
"count": 3,
"filters_applied": {
"neighborhood": "North Ames",
"min_bedrooms": 3
},
"comparables": [
{
"property_id": "PROP-0604",
"hybrid_score": 0.8945,
"semantic_score": 0.8872,
"keyword_score": 0.9034,
"comparability_score": 0.8286,
"prioritized_score": 0.8615,
"comparability_details": {
"location_similarity": 1.0,
"sqft_similarity": 0.89,
"quality_similarity": 0.89,
"age_similarity": 0.82,
"room_similarity": 0.88,
"sqft_delta": -144.0,
"price_per_sqft": 107.26
},
"data": {
```


4. Property Classification Agent

• **Endpoint**: `/api/v1/classify`

• **HTTP Method**: `POST`

• **Description**: Classifies properties into Market Tiers, Architectural Typologies, Condition Grades, and Investment Profiles.

• **Request Format**:


```
{
"property_features": {
"neighborhood": "Stone Brook",
"gr_liv_area": 2600,
"bedrooms": 4,
"bathrooms": 3,
"year_built": 2005,
"overall_qual": 9,
```

```
"overall_cond": 7
},
"query": "What is the market tier for this luxury home?"
}
```


- **Sample Response**:


```
{
"market_tier": "Luxury / High-End",
"property_typology": "Executive Estate / Sprawling Residence",
"condition_grade": "Grade A (Pristine / Turnkey Modern)",
"investment_profile": "High-Equity Core Wealth Preservation",
"query_intent": "property_valuation_request",
"classification_rationale": "Classified as Luxury / High-End (Executive Estate / Sprawling Residence) based on 2600 sq ft
living area, quality score 9/10, built in 2005...",
"target_demographic": "Affluent buyers, move-up executives seeking premium finishes."
}
```


5. Basic Property Valuation Recommendation

- **Endpoint**: `/api/v1/valuation/basic`
- **HTTP Method**: `POST`
- **Description**: Calculates comp-adjusted fair market value, median and mean comparable sale prices, size-weighted PPSF estimate, and valuation confidence range.
- **Request Format**:


```
{
"property_features": {
"neighborhood": "North Ames",
"gr_liv_area": 1800,
"bedrooms": 3,
"bathrooms": 2,
"year_built": 1995,
"overall_qual": 7
},
"top_k_comps": 5
}
```


- **Sample Response**:


```
{
"estimated_fair_market_value": 174250.0,
"valuation_range_low": 142100.0,
"valuation_range_high": 206400.0,
"median_comparable_price": 172000.0,
"mean_comparable_price": 175300.0,
"implied_price_per_sqft": 96.81,
"comparable_count_used": 5,
"confidence_level": "High (Strong comparable alignment)",
"valuation_explanation": "Based on 5 prioritized comparable sales, the estimated fair market value is $174,250.00...",
"comparables_summary": [
{
"property_id": "PROP-0604",
"neighborhood": "North Ames",
"unadjusted_sale_price": 177625.0,
"adjusted_sale_price": 186506.25,
"living_area_sqft": 1656,
"overall_quality": 6,
"year_built": 1960,
"comparability_score": 0.8286
}
]
}
```


6. Supervised ML Price Prediction

- **Endpoint**: `/api/v1/valuation/predict-price`
- **HTTP Method**: `POST`
- **Description**: Computes statistical price prediction using a trained Gradient Boosting regressor with 90% confidence interval.
- **Request Format**:

```
{
"property_features": {
"neighborhood": "North Ames",
"gr_liv_area": 1800,
"bedrooms": 3,
"bathrooms": 2,
"year_built": 1995,
"overall_qual": 7,
"overall_cond": 6,
"garage_cars": 2,
"total_bsmt_sf": 1000
}
}
```


- **Sample Response**:


```
{
"predicted_price": 206940.44,
"lower_bound_90pct": 161093.65,
"upper_bound_90pct": 252787.23,
"price_per_sqft": 114.97,
"model_r2_score": 0.9031,
"model_used": "GradientBoosting",
"input_features": {
"OverallQual": 7.0,
"OverallCond": 6.0,
"GrLivArea": 1800.0,
"Neighborhood": "North Ames"
}
}
```


7. Property Value Risk Prediction

- **Endpoint**: `/api/v1/valuation/predict-risk`
- **HTTP Method**: `POST`
- **Description**: Evaluates multi-dimensional property value risk score (0-100) across physical age depreciation, quality-condition discrepancy, market volatility, and layout obsolescence.
- **Request Format**:


```
{
"property_features": {
"neighborhood": "North Ames",
"gr_liv_area": 1800,
"bedrooms": 3,
"year_built": 1955,
"year_remod_add": 1955,
"overall_qual": 7,
"overall_cond": 4
}
}
```


- **Sample Response**:


```
{
"overall_risk_score": 62.4,
"risk_category": "High Risk",
"risk_factors": [
"Aging infrastructure: 71 years since last remodel/major system update.",
"Deferred maintenance alert: High architectural quality (7/10) coupled with degraded condition (4/10)."
],
"dimension_scores": {
"age_depreciation_risk": 75.0,
"condition_discrepancy_risk": 80.0,
"market_volatility_risk": 42.0,
"layout_liquidity_risk": 15.0
},
"mitigation_recommendations": [
"Conduct comprehensive inspection of roof, HVAC, electrical wiring, and plumbing.",
"Appraise immediate cosmetic and structural repair costs to avoid further value erosion."
],
"summary": "Overall Risk Profile assessed as 'High Risk' (Score: 62.4/100)..."
}
```

8. Feature & Location Impact Analysis

- **Endpoint**: `/api/v1/valuation/impact-analysis`
- **HTTP Method**: `POST`
- **Description**: Analyzes neighborhood price premiums and simulates what-if renovation/expansion scenarios.
- **Request Format**:


```
{
"base_features": {
"neighborhood": "North Ames",
"gr_liv_area": 1600,
"bedrooms": 3,
"overall_qual": 6,
"year_built": 1990
},
"modifications": {
"add_sqft": 300,
"upgrade_quality": 1,
"remodel": true
}
}
```


- **Sample Response**:


```
{
"simulation": {
"baseline_price": 172400.0,
"modified_price": 218500.0,
"dollar_value_added": 46100.0,
"percentage_increase": 26.74,
"modifications_applied": {
"add_sqft": 300,
"upgrade_quality": 1,
"remodel": true
}
},
"location_impact": {
"neighborhood": "North Ames",
"found": true,
"location_premium_pct": -12.5,
"dollar_premium_vs_city": -20000.0,
"median_neighborhood_price": 140000.0,
"city_baseline_median": 160000.0
},
"feature_importance": {
"top_drivers": {
"OverallQual": 0.54,
"GrLivArea": 0.22,
"TotalBsmtSF": 0.08
```


9. LLM-as-Judge Valuation Validation

- **Endpoint**: `/api/v1/valuation/judge-validate`
- **HTTP Method**: `POST`
- **Description**: Independent audit of valuation by appraisal judge agent evaluating market alignment, ML consistency, and risk coherence. Uses live LLM client when `OPENAI_API_KEY` is present, with deterministic rubric evaluation as fallback.
- **Validation**: Enforces strict input validation on `property_features` (HTTP `400 Bad Request` on invalid/out-of-bounds fields).
- **Request Format**:


```
{
"property_features": {
"neighborhood": "North Ames",
"gr_liv_area": 1800,
"overall_qual": 7,
"total_rooms": 7,
"garage_yr_blt": 1995
},
"recommended_price": 185000.0,
"ml_predicted_price": 190000.0,
"comp_median_price": 182000.0,
"comp_count": 5,
"risk_score": 28.0,
```

```
"valuation_explanation": "Comp-adjusted market estimate."
}
```


- **Sample Response**:


```
{
"verdict": "APPROVED",
"confidence_score": 0.945,
"rubric_scores": {
"market_comp_alignment": 1.0,
"ml_model_consistency": 1.0,
"risk_coherence": 0.95,
"logical_integrity": 0.95
},
"audit_commentary": "AUDIT VERDICT: APPROVED. Proposed valuation of $185,000.00 is well-grounded in market comps (median: $182,000.00) and conforms within 2.6% of ML predictive regression ($190,000.00).",
"recommended_adjustment": null,
"flags": []
}
```


10. Multi-Agent Valuation Collaboration

- **Endpoint**: `/api/v1/valuation/collaborate`
- **HTTP Method**: `POST`
- **Description**: Orchestrates the complete collaborative workflow across all agents and services into a unified Valuation Dossier.
- **Validation**: Rejects invalid property features with HTTP `400 Bad Request`.
- **Request Format**:


```
{
"property_features": {
"neighborhood": "North Ames",
"gr_liv_area": 1800,
"bedrooms": 3,
"bathrooms": 2,
"year_built": 1995,
"overall_qual": 7,
"overall_cond": 6,
"garage_cars": 2,
"total_rooms": 7,
"garage_yr_blt": 1995
},
"query": "Valuation of 3 bedroom home in North Ames built 1995",
"top_k_comps": 5
}
```


- **Sample Response**:


```
{
"is_valid": true,
"reconciled_fair_market_value": 187552.45,
"applied_calibration_factor": 1.0045,
"confidence_interval": {
"low": 128363.13,
"high": 252787.23
},
"classification": {
"market_tier": "Upper-Middle Market",
"property_typology": "Suburban Single-Family Family Residence",
"condition_grade": "Grade C (Functional Average)",
"investment_profile": "Core Suburban Appreciation"
},
"ml_prediction": {
"predicted_price": 206940.44,
"model_r2_score": 0.9031
},
"basic_valuation": {
"estimated_fair_market_value": 168164.46,
"median_comparable_price": 164000.0
},
"risk_assessment": {
"overall_risk_score": 28.8,
"risk_category": "Low Risk"
```

11. Feedback Loop: Submit Appraisal Outcome

- **Endpoint**: `/api/v1/feedback/submit`
- **HTTP Method**: `POST`
- **Description**: Submits professional appraisal feedback, verified transaction prices, ratings, and optional `applied_calibration_factor` (default 1.0) so systematic error is evaluated against the uncalibrated model estimate without drifting.
- **Request Format**:

```
{
"property_id": "PROP-0604",
"recommended_value": 172000.0,
"actual_valuation": 177625.0,
"applied_calibration_factor": 1.0,
"user_rating": 5,
"appraiser_comments": "Close alignment with closing price; minor kitchen update accounted for difference.",
"professional_role": "Certified Appraiser"
}
```

- **Sample Response**:

```
{
"feedback_id": "FB-004",
"timestamp": "2026-10-06T22:15:00",
"property_id": "PROP-0604",
"recommended_value": 172000.0,
"actual_valuation": 177625.0,
"applied_calibration_factor": 1.0,
"uncalibrated_recommended_value": 172000.0,
"error_dollars": 5625.0,
"error_pct": 3.27,
"user_rating": 5,
"appraiser_comments": "Close alignment with closing price...",
"professional_role": "Certified Appraiser"
}
```

12. Feedback Loop: Summary & Calibration Bias

- **Endpoint**: `/api/v1/feedback/summary`
- **HTTP Method**: `GET`
- **Description**: Returns continuous improvement learning metrics, historical MAE, MAPE, and systematic bias calibration factor.
- **Request Format**: None
- **Sample Response**:

```
{
"total_feedback_count": 4,
"average_rating": 4.75,
"mean_absolute_error_dollars": 5031.25,
"mean_absolute_percentage_error": 2.76,
"systematic_bias_pct": 0.45,
"calibration_factor": 1.0045,
"recent_records": [...]
}
```

13. DeepEval Quality Benchmark Execution

- **Endpoint**: `/api/v1/evaluation/run`
- **HTTP Method**: `GET`
- **Description**: Executes the DeepEval recommendation quality benchmark test suite across 5 curated test scenarios. Response depends on whether `OPENAI_API_KEY` is configured.

Offline mode (no API key) — `evaluation_mode = "offline_no_llm"`:

```
{
"evaluation_framework": "DeepEval (v4.2.8)",
"evaluation_mode": "offline_no_llm",
"execution_time_seconds": 4.1,
"total_scenarios_evaluated": 5,
"passed_scenarios": null,
"pass_rate_percentage": null,
```

```
"mean_scores": {
"mean_faithfulness": null,
"mean_answer_relevancy": null,
"mean_contextual_precision": null,
"mean_contextual_recall": null,
"mean_valuation_accuracy": 0.893
},
"benchmark_status": "NOT_EVALUATED"
}
```


The four LLM-backed DeepEval metrics are skipped (`deepeval_metric_status: "SKIPPED_NO_LLM_KEY"` per scenario).

Live mode (`OPENAI_API_KEY` set) — `evaluation_mode = "live_llm_judge"`:


```
{
"evaluation_framework": "DeepEval (v4.2.8)",
"evaluation_mode": "live_llm_judge",
"execution_time_seconds": 18.4,
"total_scenarios_evaluated": 5,
"passed_scenarios": 4,
"pass_rate_percentage": 80.0,
"mean_scores": {
"mean_faithfulness": 0.82,
"mean_answer_relevancy": 0.79,
"mean_contextual_precision": 0.76,
"mean_contextual_recall": 0.73,
"mean_valuation_accuracy": 0.893
},
"benchmark_status": "PASSED"
}
```


> Note: Live score values are illustrative. Actual scores depend on the LLM provider and run-time.

14. Dataset Information & Summary

- **Endpoint**: `/api/v1/dataset/info`
- **HTTP Method**: `GET`
- **Description**: Returns dataset documentation and statistical properties for the Ames Housing dataset.
- **Request Format**: None
- **Sample Response**:


```
{
"dataset_name": "Ames Housing Dataset",
"source": "OpenML Data ID 43926 / Dean De Cock (2011)",
"total_records": 2930,
"key_fields": [
"SalePrice", "OverallQual", "OverallCond", "GrLivArea",
"TotalBsmtSF", "GarageCars", "GarageArea", "YearBuilt",
"YearRemodAdd", "FullBath", "BedroomAbvGr", "TotRmsAbvGrd",
"Neighborhood", "LotArea", "GarageYrBlt"
],
"neighborhoods_count": 28,
"price_statistics": {
"min": 12789.0,
"median": 160000.0,
"mean": 180796.06,
"max": 755000.0
}
}
```